

1.What is need for message authrntication? List various techniques used for message authentication. Explain any one.

Need for Message Authentication

Message Authentication is required to ensure that the **message received is genuine, not altered, and sent by an authorized sender.**

Need / Importance:

1. **Integrity** – Ensures message is not modified during transmission
2. **Authentication** – Verifies the identity of the sender
3. **Prevention of Forgery** – Stops unauthorized users from sending fake messages
4. **Protection against Replay Attacks** – Prevents reuse of old messages
5. **Trustworthy Communication** – Ensures secure data exchange

Techniques for Message Authentication

1. **Message Authentication Code (MAC)**
2. **Hash Functions (MD5, SHA)**
3. **Digital Signatures**
4. **Symmetric Key Authentication**
5. **Public Key Authentication**

Digital Signature

<https://www.geeksforgeeks.org/computer-networks/digital-signatures-certificates/>

A digital signature is a cryptographic technique used to verify the authenticity, integrity, and non-repudiation of a digital message or document. It ensures that the message was created by a known sender and that it has not been altered during transmission.

Definition:

A digital signature is a cryptographic technique used to verify the authenticity and integrity of a message using public key cryptography.

Working:

Sender:

- **Creates hash of the message**
- **Encrypts hash using private key (digital signature)**
- **Sends message and digital signature**

Receiver:

- Decrypts signature using public key
- Compares with computed hash
- If both match, message is authentic

Advantages:

- Provides authentication
- Ensures data integrity
- Supports non-repudiation

Disadvantages:

- More complex than MAC
- Requires key management

